



Analysis of Fully Discrete Approximations for Dissipative Systems and Application to Time-Dependent Nonlocal Diffusion Problems

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Abstract

In this paper we first present stability and error analysis of the fully discrete numerical schemes for general dissipative systems, in which the implicit Runge–Kutta (IRK) method is adopted for time integration. Under suitable conditions on the IRK time stepping method that we refer as the total stability, a priori error estimates can be simultaneously obtained. Then we apply such time-marching techniques and analysis framework to one-dimensional time-dependent nonlocal diffusion problems, together with the discontinuous Galerkin method being used for spatial discretization. Unconditional stability of approximations of both primal and auxiliary variables and the priori error estimates for the corresponding fully discrete systems are proved, and the results indicate the schemes are asymptotically compatible. In addition, long time asymptotic behavior of the approximate solutions is also investigated. Various numerical experiments are finally performed to verify the theoretical results.

Keywords Fully discrete · Strong stability · Implicit Runge–Kutta · Nonlocal diffusion · Asymptotic compatibility

Mathematics Subject Classification 65M60 · 65R20 · 65L06 · 65L07 · 45A05

1 Introduction

Many numerical studies of time-dependent partial differential equation (PDE) models are based on semi-discrete framework, with discretization in space only while remaining time continuous. Among different time discretization methods, Runge–Kutta (RK) methods are

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avored by a large number of people. One of the major interests in using RK methods for discrete time integration is the strong stability preserving property for the coercive problems. Based on the energy method, Levy et al. obtained in [29] the strong stability of third order and fourth order RK methods for well-posed semi-discrete approximations under a suitable CFL condition. Later in [23,34], the authors provided a simpler proof of strong stability of general s -stage s th order Runge–Kutta schemes for coercive problems, and the time step constraint was significantly relaxed. However, Many problems, such as the purely hyperbolic problems, do not satisfy the coercivity condition, and we may only have the semi-negative condition (see e.g. [33]). In [34], Tadmor removed the restriction of coercivity condition and proved the strong stability of third order RK approximations for the semi-negative problems. Sun et al. in [33] considered the fourth order RK method for the well-posed linear PDE system where the spatial operator is semi-negative, and obtained the strong stability in two steps under certain time step constraints. Moreover, there exist some works analyzing fully discrete numerical schemes on some specific models. We briefly review some earlier works on discrete time integrators for dynamic systems generated from semi-discrete in-space approximations of time-dependent models, such as the semi-discrete DG discretizations for hyperbolic conservation laws [7–10]. From 2004, Zhang et al. studied error estimates of the fully discrete Runge–Kutta discontinuous Galerkin (RKDG) method for scalar conservation laws and hyperbolic systems in [28,45]. Later in 2010, Zhang and Shu studied the stability and error estimates of the explicit third order RKDG method for scalar conservation laws by energy technique [46]. Meanwhile, there also exist some results of fully discrete analysis for other type of partial differential equations. For instance, Epshteyn et al. analyzed fully discrete DG schemes for the two-dimensional Keller–Segel chemotaxis model in [19]. Wang et al. [39,40] considered the LDG method with implicit-explicit time-marching for convection–diffusion problems and obtained the stability results and error estimates. Hochbruck and Pažur [25] considered implicit Runge–Kutta methods for the time integration of linear Maxwell’s equations, with a discontinuous Galerkin discretization in space. Cheng and Zhang [4,5] considered LDG method with explicit third order RK method for singularly perturbed problems.

In recent years the nonlocal models have attracted much attention due to their many applications in mechanics and material science [31,32], random walk model in biological modeling [18], image analyses [22], machine learning [30] and phase transitions [1,17,20], etc. So far there have been a number of theoretical and numerical investigations on nonlocal diffusion (ND) models, we refer to [13–15,17,35–38] and references cited therein. Since the solutions of the ND models could contain discontinuities, it is quite natural to adopt discontinuous Galerkin (DG) discretization in space, whose basis functions are discontinuous across the element interfaces. In [14], we developed and studied a semi-discrete DG method for solving one-dimensional time-dependent nonlocal diffusion problems. Specifically, inspired by the local DG (LDG) approach [11], we introduced an auxiliary variable and rewrote the ND equation into a new system. Note that the so-called “alternating principle” cannot be applied directly to this method since the fluxes are not explicit shown in the DG schemes for nonlocal models. With the mesh size being fixed, the proposed DG method would become the traditional LDG method for the corresponding local partial differential equation problem when the horizon goes to zero. Theoretical analysis given in [14] shows that the semi-discrete DG scheme is stable, convergent and asymptotically compatible. However, the time discretization is still an important issue that should be carefully treated; for instance, in the numerical tests presented in [14], the explicit second-order total variation diminishing (TVD) Runge–Kutta method for time stepping was used together with the semi-discrete DG discretization

in space, but the simulation results showed that the time step size was severely restricted in the produced fully discrete system for the sake of numerical stability.

In this paper, we first analyze the fully discrete numerical schemes for dissipative systems, with the implicit Runge–Kutta (IRK) method being used for time marching, and then apply such time-marching techniques and analysis framework to time-dependent nonlocal diffusion problems with semi-discretization in space by the discontinuous Galerkin (DG) method. The paper is organized as follows. In Sect. 2, we first present a family of IRK time-stepping schemes, and the well-known conditions that guarantee unconditional strong stability (i.e., without any time step size restrictions) of fully discrete schemes for dissipative systems. We then introduce a class of totally stable IRK schemes that satisfy some extra conditions in order to yield stability result not only on the numerical solution but also the solutions at intermediate stages in IRK. The latter turns out to be crucial for deriving the priori error estimates. As examples to illustrate the proposed conditions, two specific IRK methods, the singly diagonal IRK (SDIRK) method and the Gauss–Legendre (GL) method, are considered in details. Then we make some assumptions on the consistency errors of the spatial discretizations to get the error estimates of fully discrete schemes for general dissipative systems. In Sect. 3, we consider an application of such algorithmic and analysis framework to time-dependent ND models in one dimension. A fully discrete scheme is formed by combining the IRK method for time integration with the DG method developed in [14] for spatial discretization. In particular, we prove the stability and derive a priori error estimates for the resulting fully discrete system, and the results imply that our fully discrete numerical scheme is asymptotically compatible. Furthermore, we also prove the long time asymptotic behavior of the approximate solutions, namely, the numerical solutions would eventually converge to a constant along the time if there is no external forcing to the system. In Sect. 4, we carry out numerical experiments to verify the theoretical results. Some concluding remarks are finally given in Sect. 5.

2 IRK Time Marching Schemes for Dissipative Systems

In this section, let us consider a time-dependent PDE of the following form:

$$u_t = F(u), \quad (x, t) \in \Omega_T, \quad (2.1)$$

with the initial condition and the periodic (or other appropriate) boundary condition, where $u(x, t)$ is the unknown function, $\Omega_T = \Omega \times (0, T)$ with $\Omega \subset \mathbb{R}^n$ being open and bounded and $F(\cdot)$ is a dissipative spatial operator satisfying the dissipative condition

$$(F(u), u) = \int_{\Omega} u F(u) dx \leq 0.$$

Note that $F(\cdot)$ could be either linear or nonlinear. Let V_h denote a discrete finite dimensional solution space on Ω , and assume that we have a corresponding semi-discrete (in space) approximation of (2.1):

$$(u_h)_t = F_h(u_h), \quad (x, t) \in \Omega_T, \quad (2.2)$$

with the initial condition $u_h(0) = u_h^0 = P_h(u(0))$, where $F_h(\cdot)$ represents the discretized spatial operator with respect to $F(\cdot)$, $u_h(\cdot, t) \in V_h$ denotes the semi-discrete solution and P_h is an appropriate projection operator onto V_h . Throughout this section we assume that $F_h(\cdot)$ satisfies the following *discrete dissipative condition*:

$$(F_h(v_h), v_h) \leq 0, \quad \forall v_h \in V_h. \quad (2.3)$$

Below we introduce and study the implicit J -stage Runge–Kutta method [2] for time integration of the semi-discrete dissipative system (2.2).

2.1 General Fully Discrete Scheme

The general J -stage Runge–Kutta schemes for time integration of the semi-discrete system (2.2) are defined as follows: for $n = 0, 1, 2, \dots$,

$$\begin{cases} u_h^{n,l} = u_h^n + \tau \sum_{m=1}^J a_{l,m} F_h(u_h^{n,m}), & 1 \leq l \leq J, \\ u_h^{n+1} = u_h^n + \tau \sum_{m=1}^J b_m F_h(u_h^{n,m}), \end{cases} \quad (2.4)$$

where $\tau > 0$ is the time step size. Let $c_l = \sum_{m=1}^J a_{l,m}$ holds, then the time level of $F_h(u_h^{n,m})$ is $t_n + c_m \tau$. Then we can express the Runge–Kutta methods by their Butcher tableau in Table 1, where $\mathbf{c} = (c_1, \dots, c_J)^T$, $A = (a_{l,m})_{J \times J}$, $\mathbf{b} = (b_1, b_2, \dots, b_J)^T$.

If the matrix A is not strictly lower triangular, then it is called the implicit Runge–Kutta method. First let us define the following condition on the desired IRK method [2, Theorem 357C, Chapter 3]:

(H1) – $b_i > 0$, $i = 1, \dots, J$, and the symmetric matrix M given by

$$M = \text{diag}(\mathbf{b})A + A^T \text{diag}(\mathbf{b}) - \mathbf{b}\mathbf{b}^T,$$

is positive semi-definite.

The following well-known result has been proved in [2]:

Lemma 1 (Strong stability) *If the dissipative condition (2.3) and Condition (H1) hold, then the fully discrete scheme (2.4) is of strong stability unconditionally in the L^2 -norm sense, i.e.,*

$$\|u_h^{n+1}\|_{L^2} \leq \|u_h^n\|_{L^2}, \quad \forall n \geq 0. \quad (2.5)$$

Below we present the proof of Lemma 1 from [2] for clarification.

Proof From (2.4), we have

$$2u_h^{n,l} = u_h^{n+1} + u_h^n + \tau \sum_{m=1}^J (2a_{l,m} - b_m) F_h(u_h^{n,m}), \quad 1 \leq l \leq J. \quad (2.6)$$

Table 1 Butcher tableau for general J -stage Runge–Kutta method

$\mathbf{c}$	A
	$\mathbf{b}^T$

Take the inner product of (2.6) and $\tau b_l F_h(u_h^{n,l})$, then sum it over l , we can obtain

$$\begin{aligned} 2\tau \sum_{l=1}^J b_l(u_h^{n,l}, F_h(u_h^{n,l})) &= \left(u_h^{n+1} + u_h^n, \tau \sum_{l=1}^J b_l F_h(u_h^{n,l}) \right) \\ &\quad + \tau^2 \sum_{l,m=1}^J (2a_{l,m}b_l - b_l b_m)(F_h(u_h^{n,l}), F_h(u_h^{n,m})). \end{aligned} \quad (2.7)$$

From (2.4), we have $u_h^{n+1} - u_h^n = \tau \sum_{l=1}^J b_l F_h(u_h^{n,l})$, and thus

$$\sum_{l,m=1}^J (2a_{l,m}b_l - b_l b_m)(F_h(u_h^{n,l}), F_h(u_h^{n,m})) = \int_{\Omega} \mathbf{F}_h^T \mathbf{M} \mathbf{F}_h d\mathbf{x},$$

where $\mathbf{F}_h = (F_h(u_h^{n,1}), F_h(u_h^{n,2}), \dots, F_h(u_h^{n,J}))^T$. Therefore, from (2.7) we have

$$\|u_h^{n+1}\|_{L^2}^2 - \|u_h^n\|_{L^2}^2 = 2\tau \sum_{l=1}^J b_l(u_h^{n,l}, F_h(u_h^{n,l})) - \tau^2 \int_{\Omega} \mathbf{F}_h^T \mathbf{M} \mathbf{F}_h d\mathbf{x}, \quad (2.8)$$

which finally gives us (2.5) by using (2.3) and Condition (H1). $\square$

In addition to the above one, let us also introduce the second important condition on the IRK method (see e.g. [6, 12, 25]), as stated in the following.

(H2) – There exists $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_J)$, $\lambda_l > 0$, $l = 1, \dots, J$, such that the symmetric matrix $\tilde{M}$ defined by

$$\tilde{M} = \Lambda A + A^T \Lambda$$

is positive definite.

Under the above extra condition, we can further obtain the stability of the intermediate numerical solutions.

Lemma 2 (Stability of intermediate stages) *If the dissipative condition (2.3) and Condition (H2) hold, then the solutions of intermediate stages $u_h^{n,l}$, $l = 1, \dots, J$ of the fully discrete scheme (2.4), are L^2 -stable, and specifically,*

$$\|u_h^{n,l}\|_{L^2} \leq (1 + C_*) \|u_h^n\|_{L^2}, \quad \forall l = 1, \dots, J, \quad (2.9)$$

where $C_* = \frac{2Ja_{\max}\lambda_{\max}}{\tilde{\lambda}}$ with $a_{\max} = \max_{0 \leq l, m \leq J} |a_{l,m}| > 0$, $\lambda_{\max} = \max_{0 \leq l \leq J} \lambda_l > 0$, and $\tilde{\lambda} > 0$ being the minimum eigenvalue of the symmetric positive definite matrix $\tilde{M}$.

Proof For convenience of analysis, let us denote

$$\mathbf{U}_h^{n,*} = \begin{pmatrix} u_h^{n,1} \\ u_h^{n,2} \\ \vdots \\ u_h^{n,J} \end{pmatrix}, \quad \mathbf{U}_h^n = \begin{pmatrix} u_h^n \\ \vdots \\ u_h^n \end{pmatrix}, \quad \mathbf{F}_h(\mathbf{U}_h^{n,*}) = \begin{pmatrix} F_h(u_h^{n,1}) \\ F_h(u_h^{n,2}) \\ \vdots \\ F_h(u_h^{n,J}) \end{pmatrix}.$$

Then from (2.4), it holds

$$\mathbf{U}_h^{n,*} = \mathbf{U}_h^n + \tau \mathbf{A} \mathbf{F}_h(\mathbf{U}_h^{n,*}).$$

For any $\mathbf{U} = (U_1, \dots, U_J)^T$, $\mathbf{V} = (V_1, \dots, V_J)^T$, define the inner product $\langle \mathbf{U}, \mathbf{V} \rangle = \sum_{l=1}^J \langle U_l, V_l \rangle$. Then we have

$$\langle \mathbf{A} \mathbf{U}_h^{n,*}, \mathbf{F}_h(\mathbf{U}_h^{n,*}) \rangle - \langle \mathbf{A} \mathbf{U}_h^n, \mathbf{F}_h(\mathbf{U}_h^{n,*}) \rangle = \tau \langle \mathbf{A} \mathbf{A} \mathbf{F}_h(\mathbf{U}_h^{n,*}), \mathbf{F}_h(\mathbf{U}_h^{n,*}) \rangle. \quad (2.10)$$

Since $\tilde{\mathbf{M}} = \mathbf{A} \mathbf{A} + \mathbf{A}^T \mathbf{A}$ is symmetric positive definite from Condition (H2), we obtain

$$\langle \mathbf{A} \mathbf{A} \mathbf{F}_h(\mathbf{U}_h^{n,*}), \mathbf{F}_h(\mathbf{U}_h^{n,*}) \rangle = \frac{1}{2} \langle \tilde{\mathbf{M}} \mathbf{F}_h(\mathbf{U}_h^{n,*}), \mathbf{F}_h(\mathbf{U}_h^{n,*}) \rangle \geq \frac{\tilde{\lambda}}{2} \sum_{l=1}^J \|F_h(u_h^{n,l})\|_{L^2}^2. \quad (2.11)$$

With the dissipative condition (2.3) and Cauchy–Schwarz inequality, it also holds

$$\begin{aligned} \langle \mathbf{A} \mathbf{U}_h^{n,*}, \mathbf{F}_h(\mathbf{U}_h^{n,*}) \rangle - \langle \mathbf{A} \mathbf{U}_h^n, \mathbf{F}_h(\mathbf{U}_h^{n,*}) \rangle &= \sum_{l=1}^J \lambda_l \left(\langle u_h^{n,l}, F_h(u_h^{n,l}) \rangle - \langle u_h^n, F_h(u_h^{n,l}) \rangle \right) \\ &\leq - \sum_{l=1}^J \lambda_l \langle u_h^n, F_h(u_h^{n,l}) \rangle \\ &\leq \left(\sum_{l=1}^J \lambda_l^2 \|u_h^n\|_{L^2}^2 \right)^{\frac{1}{2}} \left(\sum_{l=1}^J \|F_h(u_h^{n,l})\|_{L^2}^2 \right)^{\frac{1}{2}}. \end{aligned} \quad (2.12)$$

Plugging (2.11) and (2.12) into (2.10), we obtain

$$\begin{aligned} \tau^2 \sum_{l=1}^J \|F_h(u_h^{n,l})\|_{L^2}^2 &\leq 4\tilde{\lambda}^{-2} \sum_{l=1}^J \lambda_l^2 \|u_h^n\|_{L^2}^2 \\ &\leq 4J\lambda_{\max}^2 \tilde{\lambda}^{-2} \|u_h^n\|_{L^2}^2. \end{aligned} \quad (2.13)$$

Thus by (2.4), (2.13), triangle inequality and Cauchy–Schwarz inequality, we get

$$\begin{aligned} \|u_h^{n,l}\|_{L^2} &\leq \|u_h^n\|_{L^2} + \tau \sum_{m=1}^J |a_{l,m}| \|F_h(u_h^{n,m})\|_{L^2} \\ &\leq \|u_h^n\|_{L^2} + \left(\sum_{m=1}^J a_{l,m}^2 \right)^{\frac{1}{2}} \left(\tau^2 \sum_{m=1}^J \|F_h(u_h^{n,m})\|_{L^2}^2 \right)^{\frac{1}{2}} \\ &\leq \|u_h^n\|_{L^2} + (J a_{\max}^2)^{\frac{1}{2}} \left(4J\lambda_{\max}^2 \tilde{\lambda}^{-2} \|u_h^n\|_{L^2}^2 \right)^{\frac{1}{2}} \\ &\leq (1 + C_*) \|u_h^n\|_{L^2} \end{aligned} \quad (2.14)$$

for $l = 1, 2, \dots, J$. □

Remark 1 Though we consider the L^2 norm specifically in Lemmas 1 and 2, one may expect to extend these results to the more general norms. In fact, if we have the discrete dissipative condition in a general inner product $(\cdot, \cdot)_g$, i.e.,

$$(F_h(v_h), v_h)_g \leq 0,$$

then we can easily obtain the similar stability results under the correspondingly induced norm $\|\cdot\|_g$.

Table 2 Butcher tableaux of SDIRK(2,1), SDIRK(3,2) and SDIRK(4,3)

$\frac{1}{2} \mid \frac{1}{2}$	$\begin{array}{c c} \sigma & \sigma \quad 0 \\ \hline 1-\sigma & 1-2\sigma \quad \sigma \\ & \frac{1}{2} \quad \frac{1}{2} \end{array}$	$\begin{array}{c c} \sigma & \sigma \quad 0 \quad 0 \\ \hline \frac{1}{2} & \frac{1}{2}-\sigma \quad \sigma \quad 0 \\ 1-\sigma & 2\sigma \quad 1-4\sigma \quad \sigma \\ & \mu \quad 1-2\mu \quad \mu \end{array}$
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Table 3 Butcher tableaux of GL(2,1), GL(4,2) and GL(6,3)

$\frac{1}{2} \mid \frac{1}{2}$	$\begin{array}{c c} \frac{1}{2} - \frac{\sqrt{3}}{6} & \frac{1}{4} \quad \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \hline \frac{1}{2} + \frac{\sqrt{3}}{6} & \frac{1}{4} + \frac{\sqrt{3}}{6} \quad \frac{1}{4} \\ & \frac{1}{2} \quad \frac{1}{2} \end{array}$	$\begin{array}{c c} \frac{1}{2} - \frac{\sqrt{15}}{10} & \frac{5}{36} \quad \frac{2}{9} - \frac{\sqrt{15}}{15} \quad \frac{5}{36} - \frac{\sqrt{15}}{30} \\ \hline \frac{1}{2} & \frac{5}{36} + \frac{\sqrt{15}}{24} \quad \frac{2}{9} \quad \frac{5}{36} - \frac{\sqrt{15}}{24} \\ \frac{1}{2} + \frac{\sqrt{15}}{10} & \frac{5}{36} + \frac{\sqrt{15}}{30} \quad \frac{2}{9} + \frac{\sqrt{15}}{15} \quad \frac{5}{36} \\ & \frac{5}{18} \quad \frac{4}{9} \quad \frac{5}{18} \end{array}$
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Lemma 2 shows the stability of the numerical solutions of intermediate stages in the IRK method, which is a nice property to have. Note that Condition (H2) would further play an important role in deriving the error estimates as we show later in the paper. We thus give a formal definition below.

Definition 1 An IRK method is called *totally stable* if it satisfies both Conditions (H1) and (H2) and is referred, for brevity, as a TSIRK method.

2.2 Examples of TSIRK

In the following, we present some examples of TSIRKs, that is, IRKs that satisfy Conditions (H1) and (H2). We will take the singly diagonal implicit Runge–Kutta (SDIRK) method [24,26] and the Gauss–Legendre (GL) method [2] as examples.

Note that the first order IRK scheme is the same as the backward Euler scheme which obviously satisfies Conditions (H1) and (H2). Thus we focus on higher order time-stepping schemes. The SDIRK schemes from the second order to fourth order are presented in Table 2. Another type of popular IRK method belongs to the Gauss–Legendre (GL) method, and their second-order, fourth-order and sixth-order GL schemes are given in Table 3. The r and J in SDIRK(r, J) or GL(r, J) mean that the method is r th order with J stages. It is easy to see that SDIRK(2,1) is the same as GL(2,1).

In particular, we set in Table 2 $\sigma = (3 + \sqrt{3})/6$ for SDIRK(3,2), and $\sigma = \cos(\pi/18)/\sqrt{3} + 1/2$, $\mu = 1/(6(2\sigma - 1)^2)$ for SDIRK(4,3). Now we verify Conditions (H1) and (H2) for the above IRK schemes. Let us take $\Lambda = 1$ for SDIRK(2,1) and GL(2,1), $\Lambda = \text{diag}(5, 1)$ for SDIRK(3,2) and GL(4,2), and $\Lambda = \text{diag}(5, 5, 1)$ for SDIRK(4,3) and GL(6,3), respectively. Note that the choice of Λ is not unique, and we just pick up one possible choice to show its existence for each scheme. Since the matrices M and $\tilde{M}$ are symmetric, we only need to check the values of their eigenvalues:

- For SDIRK(2,1) and GL(2,1), $M = 0$ and $\tilde{M} = 1$.
- For SDIRK(3,2), the eigenvalues of M are $\{(3 + 2\sqrt{3})/6, 0\}$, and the eigenvalues of $\tilde{M}$ are

$$\{7.93914744162649, 1.5249541735112637\}.$$

- For SDIRK(4,3), the eigenvalues of M are $\{1.5530334192387472, 0, 0\}$, and the eigenvalues of $\tilde{M}$ are

$$\{14.70274033227469, 7.919928155337873, 0.8860699810232766\}.$$

- For GL(4,2), the eigenvalues of M are $\{0, 0\}$, and the eigenvalues of $\tilde{M}$ are

$$\{2.557937483122504, 0.4420625168774959\}.$$

- For GL(6,3), the eigenvalues of M are $\{0, 0, 0\}$, and the eigenvalues of $\tilde{M}$ are

$$\{3.269493350827539, 0.4261799213963207, 0.19321561666502982\}.$$

Thus all the above IRK schemes satisfy Conditions **(H1)** and **(H2)**.

Remark 2 Note that not all SDIRK methods satisfy the Condition **(H1)**, even among some SDIRK methods that are labelled “strong stability preserving”. For example, some SDIRK schemes that preserve the strong stability under certain conditions were proposed in [21, 27], but they still fail to satisfy Condition **(H1)**. For instance, in [21], a SDIRK(3,2) scheme with $\sigma = \frac{(3-\sqrt{3})}{6}$ and a SDIRK(4,3) with $\sigma = 1/2 - \cos(5\pi/18)/\sqrt{3}$ are proposed. One can easily find that the eigenvalues of M for these two choices are

$$\{(3 - 2\sqrt{3})/6, 0\} \text{ and } \{-0.8007134744549882, 0.1062196269911724, \\ -0.026788818503952292\},$$

respectively, thus Condition **(H1)** is violated.

2.3 A Priori Error Estimates

Next we study a priori error estimates for the fully discrete scheme (2.4). In the remaining part of the paper we will assume that the time step size $\tau > 0$ and the mesh size $h > 0$ are sufficiently small. Within this subsection we also let $C > 0$ represent a generic constant independent of τ and h with possibly different values. Following [39], let us introduce the reference functions $u^{n,l}, u^n$ (without spatial discretization) corresponding to the IRK method:

$$\begin{cases} u^{n,l} = u^n + \tau \sum_{m=1}^J a_{l,m} F(u^{n,m}), & l = 1, \dots, J, \\ u^{n+1} = u^n + \tau \sum_{m=1}^J b_m F(u^{n,m}) + E^n, \end{cases} \quad (2.15)$$

where $u^n = u(t^n)$. Assume the IRK method is r th order and the exact solution u of (2.1) is smooth enough, then the error analysis implies that the truncation error

$$E^n = O(\tau^{r+1}) \quad (2.16)$$

and $u^{n,l}$, $l = 1, \dots, J$ are also smooth enough. Subtracting (2.15) from the corresponding scheme (2.4), gives us the following error equations:

$$\begin{cases} \bar{e}_{u,h}^{n,l} = \bar{e}_{u,h}^n + \tau \sum_{m=1}^J a_{l,m} (F(u^{n,m}) - F_h(u_h^{n,m})), & l = 1, \dots, J, \\ \bar{e}_{u,h}^{n+1} = \bar{e}_{u,h}^n + \tau \sum_{m=1}^J b_m (F(u^{n,m}) - F_h(u_h^{n,m})) + E^n, \end{cases} \quad (2.17)$$

where $\bar{e}_{u,h}^{n,l} = u^{n,l} - u_h^{n,l}$ and $\bar{e}_{u,h}^n = u^n - u_h^n$.

Let us define $f_h^{n,m} \in V_h$ such that

$$(f_h^{n,m}, v_h) = (F(u^{n,m}) - F_h(u_h^{n,m}), v_h), \quad \forall v_h \in V_h, \quad m = 1, \dots, J. \quad (2.18)$$

Take the inner products between both sides of the equations in (2.17) with the test function $v_h, w_h \in V_h$ respectively, then we obtain the following equalities:

$$(\bar{e}_{u,h}^{n,l}, v_h) = (\bar{e}_{u,h}^n, v_h) + \tau \sum_{m=1}^J a_{l,m} (f_h^{n,m}, v_h), \quad \forall v_h \in V_h, \quad l = 1, \dots, J, \quad (2.19a)$$

$$(\bar{e}_{u,h}^{n+1}, w_h) = (\bar{e}_{u,h}^n, w_h) + \tau \sum_{m=1}^J b_m (f_h^{n,m}, w_h) + (E^n, w_h), \quad \forall w_h \in V_h. \quad (2.19b)$$

Subtracting (2.19b) from twice of (2.19a) with $v_h = w_h = f_h^{n,l}$, we obtain

$$2(\bar{e}_{u,h}^{n,l}, f_h^{n,l}) = (\bar{e}_{u,h}^{n+1}, f_h^{n,l}) + \bar{e}_{u,h}^n (f_h^{n,l}, f_h^{n,l}) + \tau \sum_{m=1}^J (2a_{l,m} - b_m) (f_h^{n,m}, f_h^{n,l}) - (E^n, f_h^{n,l}). \quad (2.20)$$

Then by multiplying τb_l to both sides of (2.20) and summing it over l , it holds

$$\begin{aligned} 2\tau \sum_{l=1}^J b_l (\bar{e}_{u,h}^{n,l}, f_h^{n,l}) &= \left(\bar{e}_{u,h}^{n+1} + \bar{e}_{u,h}^n, \tau \sum_{l=1}^J b_l f_h^{n,l} \right) \\ &+ \tau^2 \sum_{l,m=1}^J (2a_{l,m} b_l - b_m b_l) (f_h^{n,m}, f_h^{n,l}) - \tau \sum_{l=1}^J b_l (E^n, f_h^{n,l}). \end{aligned} \quad (2.21)$$

Now let us decompose the error $\bar{e}_{u,h}$ into two parts (omit the superscripts without ambiguity):

$$\bar{e}_{u,h} = e_{u,h} + \varepsilon_{u,h}, \quad e_{u,h} = P_h u - u_h, \quad \varepsilon_{u,h} = u - P_h u, \quad (2.22)$$

and we also assume that there exists a constant $\beta_1 > 0$ such that

$$\|\varepsilon_{u,h}\|_{L^2} \leq Ch^{\beta_1}. \quad (2.23)$$

With the decomposition of $\bar{e}_{u,h}$ and

$$\begin{aligned} \sum_{l,m=1}^J (2a_{l,m} b_l - b_m b_l) (f_h^{n,m}, f_h^{n,l}) &= \sum_{l,m=1}^J (a_{l,m} b_l + a_{l,m} b_m - b_m b_l) (f_h^{n,m}, f_h^{n,l}) \\ &= \int_{\Omega} f_h^T M f_h dx, \end{aligned}$$

where $f_h = (f_h^{n,1}, f_h^{n,2}, \dots, f_h^{n,J})^T$ and M is the $J \times J$ matrix defined in Condition **(H1)**, we then rewrite (2.21) as follows.

$$\begin{aligned} \left(e_{u,h}^{n+1} + e_{u,h}^n, \tau \sum_{l=1}^J b_l f_h^{n,l} \right) &= -\tau^2 \int_{\Omega} f_h^T M f_h dx + 2\tau \sum_{l=1}^J b_l (e_{u,h}^{n,l}, f_h^{n,l}) \\ &\quad - \tau \sum_{l=1}^J b_l (\varepsilon_{u,h}^{n+1} - \varepsilon_{u,h}^n - 2(\varepsilon_{u,h}^{n,l} - \varepsilon_{u,h}^n), f_h^{n,l}) + \tau \sum_{l=1}^J b_l (E^n, f_h^{n,l}), \end{aligned} \quad (2.24)$$

Then take $w_h = e_{u,h}^{n+1} + e_{u,h}^n$ in (2.19b), we can obtain

$$\begin{aligned} \left(e_{u,h}^{n+1} + e_{u,h}^n, \tau \sum_{m=1}^J b_m f_h^{n,m} \right) &= \|e_{u,h}^{n+1}\|_{L^2}^2 - \|e_{u,h}^n\|_{L^2}^2 + (\varepsilon_{u,h}^{n+1} - \varepsilon_{u,h}^n, e_{u,h}^{n+1} + e_{u,h}^n) \\ &\quad - (E^n, e_{u,h}^{n+1} + e_{u,h}^n). \end{aligned} \quad (2.25)$$

Plug (2.25) into (2.24), we can obtain

$$\begin{aligned} \|e_{u,h}^{n+1}\|_{L^2}^2 - \|e_{u,h}^n\|_{L^2}^2 &= -\tau^2 \int_{\Omega} f_h^T M f_h dx + 2\tau \sum_{l=1}^J b_l (e_{u,h}^{n,l}, f_h^{n,l}) \\ &\quad - \tau \sum_{l=1}^J b_l (\varepsilon_{u,h}^{n+1} - \varepsilon_{u,h}^n - 2(\varepsilon_{u,h}^{n,l} - \varepsilon_{u,h}^n), f_h^{n,l}) + \tau \sum_{l=1}^J b_l (E^n, f_h^{n,l}) \\ &\quad - (\varepsilon_{u,h}^{n+1} - \varepsilon_{u,h}^n, e_{u,h}^{n+1} + e_{u,h}^n) + (E^n, e_{u,h}^{n+1} + e_{u,h}^n). \end{aligned} \quad (2.26)$$

With the smoothness assumption of u , from (2.23) and the linearity of the projection P_h we have

$$\begin{cases} \|\varepsilon_{u,h}^{n,l} - \varepsilon_{u,h}^n\|_{L^2} \leq C\tau h^{\beta_1}, & \forall l = 1, \dots, J, \\ \|\varepsilon_{u,h}^{n+1} - \varepsilon_{u,h}^n\|_{L^2} \leq C\tau h^{\beta_1}. \end{cases} \quad (2.27)$$

Thus, by Condition **(H1)**, (2.27) and Cauchy–Schwarz inequality we obtain

$$\begin{cases} \int_{\Omega} f_h^T M f_h dx \geq 0, \\ \sum_{l=1}^J b_l (\varepsilon_{u,h}^{n+1} - \varepsilon_{u,h}^n - 2(\varepsilon_{u,h}^{n,l} - \varepsilon_{u,h}^n), f_h^{n,l}) \leq Ch^{2\beta_1} + \frac{\tau^2}{2C_1} \sum_{l=1}^J \|f_h^{n,l}\|_{L^2}^2, \\ (\varepsilon_{u,h}^{n+1} - \varepsilon_{u,h}^n, e_{u,h}^{n+1} + e_{u,h}^n) \leq C\tau h^{2\beta_1} + \frac{\tau}{2} \|e_{u,h}^{n+1}\|_{L^2}^2 + \frac{\tau}{4} \|e_{u,h}^n\|_{L^2}^2, \end{cases} \quad (2.28)$$

where $C_1 > 0$ is a constant independent of τ and h , and will be explicitly determined later. Also, by (2.16) and Cauchy–Schwarz inequality, we have

$$\begin{cases} (E^n, e_{u,h}^{n+1} + e_{u,h}^n) \leq C\tau^{2r+1} + \frac{\tau}{2} \|e_{u,h}^{n+1}\|_{L^2}^2 + \frac{\tau}{4} \|e_{u,h}^n\|_{L^2}^2, \\ \sum_{l=1}^J b_l (E^n, f_h^{n,l}) \leq C\tau^{2r} + \frac{\tau^2}{2C_1} \sum_{l=1}^J \|f_h^{n,l}\|_{L^2}^2, \end{cases} \quad (2.29)$$

where r is the order of the IRK method. After plugging (2.28) and (2.29) into (2.26), we then obtain: for $n = 0, 1, 2, \dots$,

$$(1 - \tau) \|e_{u,h}^{n+1}\|_{L^2}^2 - \left(1 + \frac{\tau}{2}\right) \|e_{u,h}^n\|_{L^2}^2 \leq C\tau h^{2\beta_1} + C\tau^{2r+1} + \Theta, \quad (2.30)$$

where the term Θ is given by

$$\Theta = 2\tau \sum_{l=1}^J b_l(e_{u,h}^{n,l}, f_h^{n,l}) + \frac{\tau^3}{C_1} \sum_{l=1}^J \|f_h^{n,l}\|_{L^2}^2. \quad (2.31)$$

Note that $\|e_{u,h}^0\|_{L^2} = 0$ since $P_h(u^0) = u_h^0$.

Theorem 1 Assume that the discrete dissipative condition (2.3) holds and the exact solution u of the model problem (2.1) is smooth enough. Let $u_h^n \in V_h$ be the approximate solution obtained from the fully discrete scheme (2.4) using a r th order TSIRK time discretization. If (2.23) holds and there exist $\beta_2, \beta_3 > 0$ independent of τ and h such that

$$(e_{u,h}^{n,l}, f_h^{n,l}) \leq Ch^{\beta_2} \|e_{u,h}^{n,l}\|_{L^2} + Ch^{2\beta_3}, \quad l = 1, \dots, J, \quad (2.32)$$

then we have the following error estimate:

$$\max_{n \leq T} \|u(\cdot, t_n) - u_h^n\|_{L^2} \leq C(h^{\beta_0} + \tau^r), \quad (2.33)$$

where $\beta_0 = \min\{\beta_1, \beta_2, \beta_3\}$.

Proof Taking $v_h = f_h^{n,l}$ in (2.19a) gives us

$$(\bar{e}_{u,h}^{n,l} - \bar{e}_{u,h}^n, f_h^{n,l}) = \tau \sum_{m=1}^J a_{l,m}(f_h^{n,m}, f_h^{n,l}). \quad (2.34)$$

By multiplying λ_l with (2.34) and summing it over l , we get

$$\sum_{l=1}^J \lambda_l (\bar{e}_{u,h}^{n,l} - \bar{e}_{u,h}^n, f_h^{n,l}) = \tau \sum_{l,m=1}^J \lambda_l a_{l,m}(f_h^{n,m}, f_h^{n,l}), \quad (2.35)$$

where λ_l is defined in Condition (H2). Also, by Condition (H2), we have

$$\sum_{l,m=1}^J \lambda_l a_{l,m}(f_h^{n,m}, f_h^{n,l}) \geq \frac{\tilde{\lambda}}{2} \sum_{l=1}^J \|f_h^{n,l}\|_{L^2}^2, \quad (2.36)$$

where $\tilde{\lambda} > 0$ denotes the minimum eigenvalue of $\tilde{M}$ as defined in Lemma 2. By (2.27), (2.32) and Cauchy–Schwarz inequality, we have

$$\begin{aligned} \sum_{l=1}^J \lambda_l (\bar{e}_{u,h}^{n,l} - \bar{e}_{u,h}^n, f_h^{n,l}) &= \sum_{l=1}^J \lambda_l (e_{u,h}^{n,l}, f_h^{n,l}) - \sum_{l=1}^J \lambda_l (e_{u,h}^n, f_h^{n,l}) + \sum_{l=1}^J \lambda_l (\varepsilon_{u,h}^{n,l} \\ &\quad - \varepsilon_{u,h}^n, f_h^{n,l}) \leq Ch^{\beta_2} \sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2} + Ch^{2\beta_3} + \left(\sum_{l=1}^J \|f_h^{n,l}\|_{L^2}^2 \right)^{\frac{1}{2}} \left(\sum_{l=1}^J \lambda_l^2 \|e_{u,h}^{n,l}\|_{L^2}^2 \right)^{\frac{1}{2}} \\ &\quad + C\tau h^{\beta_1} \left(\sum_{l=1}^J \|f_h^{n,l}\|_{L^2}^2 \right)^{\frac{1}{2}}. \end{aligned} \quad (2.37)$$

Plugging (2.36) and (2.37) into (2.35), we get

$$\frac{\tilde{\lambda}}{2}\xi^2 - d_1\xi - d_2 \leq 0, \quad (2.38)$$

where

$$\begin{aligned} \xi &= \left(\tau^2 \sum_{l=1}^J \|f_h^{n,l}\|_{L^2}^2 \right)^{\frac{1}{2}}, \quad d_1 = \left(\sum_{l=1}^J \lambda_l^2 \|e_{u,h}^n\|_{L^2}^2 \right)^{\frac{1}{2}} \\ &\quad + C\tau h^{\beta_1}, \quad d_2 = C\tau h^{\beta_2} \sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2} + C\tau h^{2\beta_3}. \end{aligned}$$

From (2.38), we can further obtain

$$\xi^2 \leq \frac{2d_1\xi}{\tilde{\lambda}} + \frac{2d_2}{\tilde{\lambda}} \leq \frac{2d_1^2}{\tilde{\lambda}^2} + \frac{\xi^2}{2} + \frac{2d_2}{\tilde{\lambda}},$$

thus

$$\xi^2 \leq \frac{4d_1^2}{\tilde{\lambda}^2} + \frac{4d_2}{\tilde{\lambda}},$$

which gives us

$$\begin{aligned} \tau^2 \sum_{l=1}^J \|f_h^{n,l}\|_{L^2}^2 &\leq 4\tilde{\lambda}^{-2} \left(\left(\sum_{l=1}^J \lambda_l^2 \|e_{u,h}^n\|_{L^2}^2 \right)^{\frac{1}{2}} + C\tau h^{\beta_1} \right)^2 \\ &\quad + C\tau h^{\beta_2} \sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2} + C\tau h^{2\beta_3} \\ &\leq 8J\lambda_{\max}^2 \tilde{\lambda}^{-2} \|e_{u,h}^n\|_{L^2}^2 + C\tau^2 h^{2\beta_1} + C\tau h^{\beta_2} \sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2} + C\tau h^{2\beta_3}. \end{aligned} \quad (2.39)$$

Take $v_h = e_{u,h}^{n,l}$ in (2.19a), we obtain

$$(\bar{e}_{u,h}^{n,l} - \bar{e}_{u,h}^n, e_{u,h}^{n,l}) = \tau \sum_{m=1}^J a_{l,m} (f_h^{n,m}, e_{u,h}^{n,l}). \quad (2.40)$$

By Cauchy–Schwarz inequality, we have

$$\tau \sum_{m=1}^J a_{l,m} (f_h^{n,m}, e_{u,h}^{n,l}) \leq a_{\max} \sqrt{J} \left(\tau^2 \sum_{m=1}^J \|f_h^{n,m}\|_{L^2}^2 \right)^{\frac{1}{2}} \|e_{u,h}^{n,l}\|_{L^2}. \quad (2.41)$$

By (2.40), (2.27), (2.41) and Cauchy–Schwarz inequality, we have

$$\begin{aligned} (e_{u,h}^{n,l} - e_{u,h}^n, e_{u,h}^{n,l}) &= -(\varepsilon_{u,h}^{n,l} - \varepsilon_{u,h}^n, e_{u,h}^{n,l}) + \tau \sum_{m=1}^J a_{l,m} (f_h^{n,m}, e_{u,h}^{n,l}) \\ &\leq C\tau h^{\beta_1} \|e_{u,h}^{n,l}\|_{L^2} + a_{\max} \sqrt{J} \left(\tau^2 \sum_{m=1}^J \|f_h^{n,m}\|_{L^2}^2 \right)^{\frac{1}{2}} \|e_{u,h}^{n,l}\|_{L^2}. \end{aligned} \quad (2.42)$$

Therefore, by using (2.42) with the identity

$$(e_{u,h}^{n,l} - e_{u,h}^n, e_{u,h}^{n,l}) = \frac{1}{2} \|e_{u,h}^{n,l}\|_{L^2}^2 - \frac{1}{2} \|e_{u,h}^n\|_{L^2}^2 + \frac{1}{2} \|e_{u,h}^{n,l} - e_{u,h}^n\|_{L^2}^2,$$

we obtain that

$$\|e_{u,h}^{n,l}\|_{L^2}^2 \leq 2 \left(a_{\max} \sqrt{J} \left(\tau^2 \sum_{m=1}^J \|f_h^{n,m}\|_{L^2}^2 \right)^{\frac{1}{2}} + C\tau h^{\beta_1} \right) \|e_{u,h}^{n,l}\|_{L^2} + \|e_{u,h}^n\|_{L^2}^2, \quad (2.43)$$

and consequently by Cauchy–Schwarz inequality it deduces

$$\|e_{u,h}^{n,l}\|_{L^2}^2 \leq 2 \left(a_{\max} \sqrt{J} \left(\tau^2 \sum_{m=1}^J \|f_h^{n,m}\|_{L^2}^2 \right)^{\frac{1}{2}} + C\tau h^{\beta_1} \right)^2 + \frac{1}{2} \|e_{u,h}^{n,l}\|_{L^2}^2 + \|e_{u,h}^n\|_{L^2}^2 \quad (2.44)$$

and then

$$\begin{aligned} \|e_{u,h}^{n,l}\|_{L^2}^2 &\leq 4 \left(a_{\max} \sqrt{J} \left(\tau^2 \sum_{m=1}^J \|f_h^{n,m}\|_{L^2}^2 \right)^{\frac{1}{2}} + C\tau h^{\beta_1} \right)^2 + 2\|e_{u,h}^n\|_{L^2}^2 \\ &\leq 8a_{\max}^2 J \left(\tau^2 \sum_{m=1}^J \|f_h^{n,m}\|_{L^2}^2 \right) + C\tau^2 h^{2\beta_1} + 2\|e_{u,h}^n\|_{L^2}^2. \end{aligned} \quad (2.45)$$

Plugging (2.39) into (2.45), and summing it over l , we obtain that

$$\begin{aligned} \sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2}^2 &\leq (2 + 16C_*^2)J \|e_{u,h}^n\|_{L^2}^2 + C\tau h^{\beta_2} \sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2} + C\tau^2 h^{2\beta_1} + C\tau h^{2\beta_3} \\ &\leq \frac{1}{2} \sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2}^2 + (2 + 16C_*^2)J \|e_{u,h}^n\|_{L^2}^2 + C\tau^2 h^{2\beta_1} + C\tau^2 h^{2\beta_2} \\ &\quad + C\tau h^{2\beta_3}. \end{aligned} \quad (2.46)$$

Thus we have by (2.46) that

$$\sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2}^2 \leq C_2 \|e_{u,h}^n\|_{L^2}^2 + C\tau^2 h^{2\beta_1} + C\tau^2 h^{2\beta_2} + C\tau h^{2\beta_3}, \quad (2.47)$$

where $C_2 = 4(1 + 8C_*^2)J$.

With (2.32), (2.47) and Cauchy-Schwarz inequality, we obtain the estimate of the first term of Θ :

$$\begin{aligned} 2\tau \sum_{l=1}^J b_l(e_{u,h}^{n,l}, f_h^{n,l}) &\leq C\tau h^{\beta_2} \sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2} + C\tau h^{2\beta_3} \\ &\leq \frac{\tau}{4C_2} \sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2}^2 + C\tau h^{2\beta_2} + C\tau h^{2\beta_3} \\ &\leq \frac{\tau}{4} \|e_{u,h}^n\|_{L^2}^2 + C\tau^3 h^{2\beta_1} + C\tau h^{2\beta_2} + C\tau h^{2\beta_3}. \end{aligned} \quad (2.48)$$

With (2.39), (2.47) and Cauchy-Schwarz inequality, we obtain

$$\begin{aligned} \tau^2 \sum_{l=1}^J \|f_h^{n,l}\|_{L^2}^2 &\leq 8J\lambda_{\max}^2 \tilde{\lambda}^{-2} \|e_{u,h}^n\|_{L^2}^2 + \sum_{l=1}^J \|e_{u,h}^{n,l}\|_{L^2}^2 + C\tau^2 h^{2\beta_1} + C\tau^2 h^{2\beta_2} + C\tau h^{2\beta_3} \\ &\leq C_3 \|e_{u,h}^n\|_{L^2}^2 + C\tau^2 h^{2\beta_1} + C\tau^2 h^{2\beta_2} + C\tau h^{2\beta_3}, \end{aligned} \quad (2.49)$$

where $C_3 = 8J\lambda_{\max}^2 \tilde{\lambda}^{-2} + C_2$. By taking $C_1 = 4C_3$ and plugging (2.48) and (2.49) into (2.31), we get

$$\Theta \leq \frac{\tau}{2} \|e_{u,h}^n\|_{L^2}^2 + C\tau^3 h^{2\beta_1} + C\tau h^{2\beta_2} + C\tau h^{2\beta_3}. \quad (2.50)$$

Finally, by plugging (2.50) into (2.30) and using the discrete Gronwall's inequality and the triangle inequality, we obtain the error estimate (2.33). $\square$

3 Application to Nonlocal Diffusion Problems

In this section, we first briefly present the one-dimensional time-dependent nonlocal diffusion (ND) problem and its semi-discrete DG scheme proposed in [14], which satisfies the discrete dissipative condition (2.3). Then we propose the fully discrete scheme by further adopting the TSIRK time stepping method satisfying conditions (H1) and (H2). Based on the results from the preceding section, extensive stability analysis and a priori error estimates of the scheme will be rigorously studied, together with the long time behavior of the numerical solutions.

3.1 The Nonlocal Diffusion Problem and Its Full Discretization

We consider the one-dimensional time-dependent nonlocal diffusion (ND) problem as follows:

$$\begin{cases} u_t + \mathcal{L}_\delta u = 0, & (x, t) \in \Omega_T, \\ u(x, 0) = u_0(x), & x \in \Omega \end{cases} \quad (3.1)$$

with the periodic boundary condition, where $\Omega = (a, b)$, $\Omega_T = \Omega \times (0, T)$ and the linear operator $\mathcal{L}_\delta$ is defined as

$$\mathcal{L}_\delta u(x, t) := -2 \int_{x-\delta}^{x+\delta} (u(y, t) - u(x, t)) \widehat{\gamma}(x, y) dy. \quad (3.2)$$

The kernel function $\widehat{\gamma}(x, y)$ is assumed to satisfy the following conditions:

$$\begin{cases} \widehat{\gamma}(x, y) = \widehat{\gamma}(y, x) \geq 0, \\ \widehat{\gamma}(x, y) = \gamma(|x - y|) = \gamma(s), \quad s = |x - y|, \\ |x - y|^2 \widehat{\gamma}(x - y) \in L^1_{loc}(\mathbb{R}), \end{cases}$$

where $\gamma(s)$ is a single-variable function. With the change of variables, we obtain

$$\mathcal{L}_\delta u(x, t) = -2 \int_{-\delta}^{\delta} \gamma(s)(u(x + s, t) - u(x, t)) ds. \quad (3.3)$$

With the properties of $\gamma(s)$, we immediately obtain

$$(-\mathcal{L}_\delta u, u) = - \int_{-\delta}^{\delta} \gamma(s) \int_a^b (u(x + s, t) - u(x, t))^2 dx ds \leq 0.$$

which implies that the linear operator $Fu = -\mathcal{L}_\delta u$ in the form of (2.1) is dissipative and the L^2 -norm of the solution u is nonincreasing against time.

Since $s^2\gamma(s) \in L^1_{loc}(\mathbb{R})$, then we have $C_\delta = \int_{-\delta}^{\delta} s^2\gamma(s) ds < \infty$. Let us further assume that

$$C_\delta \rightarrow 1, \quad \text{as } \delta \rightarrow 0.$$

Thus when $\delta \rightarrow 0$, the ND problem (3.1) becomes the classic heat equation. For more details about the ND model we refer to [13].

To construct the DG scheme for the problem (3.1), let us first divide the spatial domain Ω into N cells, i.e.,

$$\mathcal{T}_h = \left\{ I_j = \left(x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}} \right) \right\}_{j=1}^N, \quad a = x_{\frac{1}{2}} < x_{\frac{3}{2}} < \cdots < x_{N+\frac{1}{2}} = b.$$

Denote $h_j = x_{j+\frac{1}{2}} - x_{j-\frac{1}{2}}$. The partition $\mathcal{T}_h$ is assumed to be regular. V_h is the finite element space defined as

$$V_h = V_h^k = \{v \in L^2 : v \in \mathcal{P}_k(I_j), j = 1, \dots, N. \text{ } v \text{ is periodic} \}, \quad (3.4)$$

where $\mathcal{P}_k(I_j)$ is a space of polynomials on I_j whose degrees are at most k . The mesh-dependent broken Sobolev space is defined as

$$H^1(\mathcal{T}_h) = \{v \in L^2((a, b)) : v|_{I_j} \in H^1(I_j), \quad \forall j = 1, 2, \dots, N\}.$$

It holds $V_h \subset H^1(\mathcal{T}_h)$.

In [14], an auxiliary variable $q(x, t; s)$ was introduced and used to develop a semi-discrete DG scheme for (3.1) as follows: find $u_h \in V_h$ such that

$$((u_h)_t, v_h) - 2 \int_0^\delta s^2 \gamma(s) \mathcal{K}(q_h, v_h; s) ds = 0, \quad \forall v_h \in V_h, \quad (3.5)$$

with $q_h(\cdot, t; s) \in V_h$ satisfying

$$(q_h, w_h) = \mathcal{H}(u_h, w_h; s), \quad \forall w_h \in V_h, \quad (3.6)$$

where the operators $\mathcal{H}$ and $\mathcal{K}$ are defined by

$$\begin{aligned}\mathcal{H}(v, w; s) &= \sum_j \mathcal{H}_j(v, w; s), \quad \mathcal{H}_j(v, w; s) = \int_{I_j} \frac{1}{s} (v(x+s) - v(x)) w(x) dx, \\ \mathcal{K}(v, w; s) &= \sum_j \mathcal{K}_j(v, w; s), \quad \mathcal{K}_j(v, w; s) = \int_{I_j} \frac{1}{s} (v(x) - v(x-s)) w(x) dx.\end{aligned}$$

It is easy to see that q_h on I_j is locally solvable.

The scheme (3.5) is analogous to the local DG (LDG) method [11], in which auxiliary variables are introduced to approximate the low-order derivatives. The main idea of the LDG method is to introduce the auxiliary variables, and then rewrite the model problem into a first order system. As the equations in the new system only involve the first order derivatives, the technique of treating the convective terms in the DG methods can be applied. In fact, the LDG method has been widely used in handling time-dependent PDEs with high order spatial derivatives, such as convection–diffusion problems, Korteweg–de Vries type equations, nonlinear Schrödinger equations, Cahn–Hilliard type equations, and biharmonic type equations, etc (e.g. see [11, 16, 41–44]). The interface numerical fluxes play an important role in ensuring the spatial stability and local solvability of the auxiliary variables. The selection of the numerical fluxes is based on some principles, such as upwind property and alternating principle, etc.

Remark 3 As LDG method works very successfully for the time-dependent PDEs containing high order spatial derivatives, it is not so straight-forward to apply it to the nonlocal diffusion problems. Since the differential operator is replaced by the integral-type operator, there are no fluxes explicitly appeared in the variational form of the nonlocal model [14]. Therefore, the alternating principle cannot be applied in designing the DG method for nonlocal problems. Fortunately, it is shown in the semi-discrete DG scheme (3.5) would become the LDG scheme when $\delta \rightarrow 0$.

Let us first present some properties of the finite element space V_h and the operators $\mathcal{H}$ and $\mathcal{K}$ in this subsection. The following lemma was proved in [14].

Lemma 3 For any $v, w \in V_h$, the following equalities hold:

$$\begin{aligned}\mathcal{H}(v, w; s) + \mathcal{K}(w, v; s) &= 0, \\ \mathcal{H}(v, v; s) &= -\frac{1}{2} \int_a^b \frac{1}{s} (v(x+s) - v(x))^2 dx, \\ \mathcal{K}(w, w; s) &= \frac{1}{2} \int_a^b \frac{1}{s} (w(x+s) - w(x))^2 dx.\end{aligned}$$

For convenience of notations, we define two bilinear operators

$$A_h(v_h, w_h) = 2 \int_0^\delta s^2 \gamma(s) \mathcal{K}(v_h, w_h; s) ds, \quad B_h(v_h, w_h) = 2 \int_0^\delta s^2 \gamma(s) \mathcal{H}(v_h, w_h; s) ds.$$

Since q_h is determined by u_h in (3.6), we denote $q_h = Q_h D_s^+ u_h$ where $D_s^+ u_h = (u_h(x+s) - u_h(x))/s$ (clearly Q_h here is the standard L^2 projection onto V_h). Based on (3.5), the discrete linear dissipative operator F_h can be correspondingly expressed in the variational form as

$$(F_h u_h, v_h) = A_h(Q_h D_s^+ u_h, v_h), \quad \forall v_h \in V_h. \quad (3.7)$$

By Lemma 3 and (3.6), we immediately obtain

$$\begin{aligned}(F_h u_h, u_h) &= 2 \int_0^\delta s^2 \gamma(s) \int_a^b \mathcal{K}(Q_h D_s^+ u_h, u_h; s) dx ds \\ &= -2 \int_0^\delta s^2 \gamma(s) \int_a^b \mathcal{H}(u_h, Q_h D_s^+ u_h; s) dx ds \\ &= -2 \int_0^\delta s^2 \gamma(s) \int_a^b (Q_h D_s^+ u_h)(x, t; s)^2 dx ds \leq 0,\end{aligned}$$

i.e., the discrete dissipative condition (2.3) is satisfied. For the semi-discrete DG scheme (3.5) we also have the following stability result for q_h proved in [14]:

$$\frac{d}{dt} \|q_h\|_S^2 \leq 0, \quad (3.8)$$

where

$$\|v\|_S^2 = (v, v)_S, (v, w)_S = 2 \int_0^\delta s^2 \gamma(s) \sum_j \int_{I_j} v(x; s) w(x; s) dx ds.$$

With the J -stage implicit Runge–Kutta method being used for time discretization, we obtain a fully discrete scheme as follows: for $n = 0, 1, 2, \dots$,

$$\begin{cases} (u_h^{n,l}, v_h) = (u_h^n, v_h) + \tau \sum_{m=1}^J a_{l,m} A_h(q_h^{n,m}, v_h), & \forall v_h \in V_h, \quad l = 1, \dots, J, \\ (u_h^{n+1}, w_h) = (u_h^n, w_h) + \tau \sum_{m=1}^J b_m A_h(q_h^{n,m}, w_h), & \forall w_h \in V_h, \end{cases} \quad (3.9)$$

where $q_h^{n,l} \in V_h$ satisfies

$$(q_h^{n,l}, z_h) = \mathcal{H}(u_h^{n,l}, z_h; s), \quad \forall z_h \in V_h, \quad l = 1, \dots, J. \quad (3.10)$$

3.2 Stability of the Fully Discrete Scheme

Since the discrete dissipative condition (2.3) holds for the semi-discrete DG scheme, we immediately obtain the stability of the approximate solution u_h for the fully discrete scheme (3.9) according to Lemmas 1 and 2. Next we show that the fully discrete scheme (3.9) also inherits a time discrete analog of the stability property (3.8) as well. Let us denote

$$(q_h^n, v_h) = \mathcal{H}(u_h^n, v_h; s), \quad \forall v_h \in V_h,$$

and introduce variables $p_h^{n,l} \in V_h$ as

$$(p_h^{n,l}, v_h) = A_h(q_h^{n,l}, v_h), \quad \forall v_h \in V_h, \quad l = 1, \dots, J. \quad (3.11)$$

Then from the scheme (3.9), we immediately obtain

$$\begin{aligned}u_h^{n,l} - u_h^n &= \tau \sum_{m=1}^J a_{l,m} p_h^{n,m}, \quad l = 1, \dots, J, \\ u_h^{n+1} - u_h^n &= \tau \sum_{m=1}^J b_m p_h^{n,m}.\end{aligned}$$

Therefore, for any $v_h, w_h \in V_h$ we have

$$\begin{cases} (q_h^{n,l} - q_h^n, v_h) = \mathcal{H}(u_h^{n,l} - u_h^n, v_h; s) = \tau \sum_{m=1}^J a_{l,m} \mathcal{H}(p_h^{n,m}, v_h; s), & l = 1, \dots, J, \\ (q_h^{n+1} - q_h^n, w_h) = \mathcal{H}(u_h^{n+1} - u_h^n, w_h; s) = \tau \sum_{m=1}^J b_m \mathcal{H}(p_h^{n,m}, w_h; s). \end{cases} \quad (3.12)$$

The fully discrete scheme (3.12) and (3.11) for q_h are very similar to the scheme (3.9) and (3.10) for u_h . In fact, by using almost the same analysis procedure as the proof in Lemma 1, we also can obtain the stability of q_h in the $\mathcal{S}$ -norm sense. A sketch proof for the stability of q_h is presented in the following.

Proof If we introduce $r_h^{n,l} \in V_h$ such that

$$(r_h^{n,l}, v_h) = \mathcal{H}(p_h^{n,l}, v_h; s), \quad l = 1, \dots, J,$$

then from (3.11) we have

$$(r_h^{n,l}, q_h^{n,l})_{\mathcal{S}} = B_h(p_h^{n,l}, q_h^{n,l}) = -A_h(q_h^{n,l}, p_h^{n,l}) = -\|p_h^{n,l}\|_{L^2}^2 \leq 0. \quad (3.13)$$

It is the dissipative condition analogue to (2.3). From (3.12) we have

$$\begin{cases} q_h^{n,l} = q_h^n + \tau \sum_{m=1}^J a_{l,m} r_h^{n,m}, & 1 \leq l \leq J, \\ q_h^{n+1} = q_h^n + \tau \sum_{m=1}^J b_m r_h^{n,m}, \end{cases} \quad (3.14)$$

Follow the similar procedure in the proof of Lemma 1, we can obtain the strong stability of q_h in $\|\cdot\|_{\mathcal{S}}$. $\square$

We summarize the results in the following theorem:

Theorem 2 For the fully discrete scheme (3.9), if Condition (H1) holds, then we have the following unconditionally strong stability results:

$$\|u_h^{n+1}\|_{L^2} \leq \|u_h^n\|_{L^2}, \quad \|q_h^{n+1}\|_{\mathcal{S}} \leq \|q_h^n\|_{\mathcal{S}}, \quad n \geq 0, \quad (3.15)$$

where u_h^n is the approximate solution of $u(x, t)$ and q_h^n is the approximate solution of $(u(x + s) - u(x))/s$ at time t_n . Furthermore, if Condition (H2) holds, then we also have the following stability results on the intermediate stages:

$$\sum_{l=1}^J \|u_h^{n,l}\|_{L^2}^2 \leq (1 + C_*) \|u_h^n\|_{L^2}^2, \quad \sum_{l=1}^J \|q_h^{n,l}\|_{\mathcal{S}}^2 \leq (1 + C_*) \|q_h^n\|_{\mathcal{S}}^2, \quad n \geq 0, \quad (3.16)$$

where $u_h^{n,l}$ and $q_h^{n,l}$ are the corresponding solutions of intermediate stages.

Remark 4 One could introduce more auxiliary variables by keep repeating this procedure as we like, for instance, if we denote

$$\begin{aligned} p_h^n &\in V_h, & (p_h^n, v_h) &= A_h(q_h^n, v_h), & \forall v_h &\in V_h, \\ g_h^n(\cdot; s) &\in V_h, & (g_h^n, w_h) &= \mathcal{H}(p_h^n, w_h; s), & \forall w_h &\in V_h, \end{aligned}$$

where p_h is the approximate solution of u_t , and g_h is the approximate solution of $(u_t(x + s) - u_t(x))/s$, then we have the following stability results:

$$\|p_h^{n+1}\|_{L^2} \leq \|p_h^n\|_{L^2}, \quad \|g_h^{n+1}\|_S \leq \|g_h^n\|_S, \quad n \geq 0. \quad (3.17)$$

3.3 A Priori Error Estimates

We assume that both Conditions **(H1)** and **(H2)** hold for the IRK method. Now we let $C > 0$ represent a generic constant independent of τ, h, δ and $\gamma(s)$ but with possibly different values throughout this subsection. We only need to show (2.23) and (2.32) hold for the scheme (3.9). We first present some basic inequalities that would be used later in the error estimate. The Gauss–Radau projection $R_h^\pm: H^1(\mathcal{T}_h) \rightarrow V_h$ is defined as: for a function $w \in H^1(\mathcal{T}_h)$,

$$\int_{I_j} (R_h^\pm w - w) v_h dx = 0, \quad \forall v_h \in \mathcal{P}^{k-1}(I_j), \quad j = 1, \dots, N. \quad (3.18)$$

and the exact collocation holds at the endpoints, i.e.,

$$R_h^+ w(x_{j-1/2}^+) = w(x_{j-1/2}^+), \quad R_h^- w(x_{j+1/2}^-) = w(x_{j+1/2}^-).$$

Denote the projection error by $\varepsilon_w = w - R_h^\pm w$, then for $w \in H^{k+1}(\Omega) \cap W_\infty^{k+1}(\Omega)$, we have the following approximation property

$$\|\varepsilon_w\|_{L^2} + \|\varepsilon_w\|_{L^\infty} \leq Ch^{k+1}, \quad (3.19)$$

where the generic constant $C > 0$ depends on k, w and its derivatives. See [3] for more information about the projection properties.

Let us adopt and specialize the error equations (2.19) to our particular setting. From the definition of $f_h^{n,l} \in V_h$ in (2.18), we have

$$(f_h^{n,l}, v_h) = A_h(\bar{e}_{q,h}^{n,l}, v_h), \quad \forall v_h \in V_h, \quad l = 1, \dots, J, \quad (3.20)$$

where

$$(\bar{e}_{q,h}^{n,l}, v_h) = \mathcal{H}(\bar{e}_{u,h}^{n,l}, v_h; s), \quad \forall v_h \in V_h, \quad l = 1, \dots, J. \quad (3.21)$$

Note $\bar{e}_{q,h}$ in the above equations are defined as follows (omit the superscripts without ambiguity):

$$\bar{e}_{q,h} = e_{q,h} + \varepsilon_{q,h}, \quad e_{q,h} = R_h^- q - q_h, \quad \varepsilon_{q,h} = q - R_h^- q.$$

Let us take $P_h = R_h^+$ in (2.22) and clearly (2.23) holds with $\beta_1 = k + 1$ due to (3.19). Take $v_h = e_{q,h}^{n,l}$ in (3.21), and from Lemma 3 we get

$$A_h(e_{q,h}^{n,l}, e_{u,h}^{n,l}) = -B_h(e_{u,h}^{n,l}, e_{q,h}^{n,l}) = B_h(\varepsilon_{u,h}^{n,l}, e_{q,h}^{n,l}) - (\bar{e}_{q,h}^{n,l}, e_{q,h}^{n,l})_S. \quad (3.22)$$

Take $v_h = e_{u,h}^{n,l}$ in (3.20), together with (3.22) we get

$$(e_{u,h}^{n,l}, f_h^{n,l}) = A_h(\varepsilon_{q,h}^{n,l}, e_{u,h}^{n,l}) + B_h(\varepsilon_{u,h}^{n,l}, e_{q,h}^{n,l}) - (e_{q,h}^{n,l}, e_{q,h}^{n,l})_S - (\varepsilon_{q,h}^{n,l}, e_{q,h}^{n,l})_S. \quad (3.23)$$

Following the analysis in [14] for the bilinear operators in space, A_h and B_h , we can easily obtain the results below:

$$A_h(\varepsilon_{q,h}^{n,l}, e_{u,h}^{n,l}) \leq \begin{cases} C(\delta)h^{k+1}\|e_{u,h}^{n,l}\|_{L^2}, & \text{for } s\gamma(s) \in L^1_{loc} \text{ and } \delta \text{ is fixed,} \\ Ch^k\|e_{u,h}^{n,l}\|_{L^2}, & \text{for general } \gamma(s) \text{ and } k \geq 1, \\ Ch^{\frac{1}{2}}\|e_{u,h}^{n,l}\|_{L^2}, & \text{for general } \gamma(s), k = 0 \text{ and } h = h_j, \end{cases} \quad (3.24)$$

$$B_h(\varepsilon_{u,h}^{n,l}, e_{q,h}^{n,l}) \leq \begin{cases} C(\delta)h^{2k+2} + \frac{1}{2}\|e_{q,h}^{n,l}\|_{\mathcal{S}}^2, & \text{for } s\gamma(s) \in L^1_{loc} \text{ and } \delta \text{ is fixed,} \\ Ch^{2k} + \frac{1}{2}\|e_{q,h}^{n,l}\|_{\mathcal{S}}^2, & \text{for general } \gamma(s) \text{ and } k \geq 1, \\ Ch + \frac{1}{2}\|e_{q,h}^{n,l}\|_{\mathcal{S}}^2, & \text{for general } \gamma(s), k = 0 \text{ and } h = h_j. \end{cases} \quad (3.25)$$

where $C(\delta) > 0$ is a constant independent of τ, h but could be dependent on δ . Note that there are slight improvements in the above estimates (3.24) and (3.25) compared with those derived in [14], specifically, the case of $s\gamma(s) \in L^1_{loc}$ with fixed δ is now included, in stead of the case of $\gamma(s) \in L^1_{loc}$ with fixed δ —the proof for this part is straightforward based on

$$\begin{aligned} A_h(\varepsilon_{q,h}^{n,l}, e_{u,h}^{n,l}) &\leq 2 \int_0^\delta s\gamma(s) \sum_j \int_{I_j} (\varepsilon_{q,h}^{n,l}(x; s) - \varepsilon_{q,h}^{n,l}(x - s; s)) e_{u,h}^{n,l}(x) dx ds \\ &\leq C(\delta)h^{k+1} \int_0^\delta s\gamma(s) \|e_{u,h}^{n,l}\|_{L^1} ds \leq C(\delta)h^{k+1} \|e_{u,h}^{n,l}\|_{L^2}, \end{aligned}$$

and a similar inequality for $B_h(\varepsilon_{u,h}^{n,l}, e_{q,h}^{n,l})$.

In addition, we have

$$(\varepsilon_{q,h}^{n,l}, e_{q,h}^{n,l})_{\mathcal{S}} \leq Ch^{2k+2} + \frac{1}{2}\|e_{q,h}^{n,l}\|_{\mathcal{S}}^2. \quad (3.26)$$

Plugging (3.24)–(3.26) into (3.23), we obtain

$$(e_{u,h}^{n,l}, f_h^{n,l}) \leq \begin{cases} C(\delta)h^{k+1}\|e_{u,h}^{n,l}\|_{L^2} + C(\delta)h^{2k+2}, & \text{for } s\gamma(s) \in L^1_{loc} \text{ and } \delta \text{ is fixed,} \\ Ch^k\|e_{u,h}^{n,l}\|_{L^2} + Ch^{2k}, & \text{for general } \gamma(s) \text{ and } k \geq 1, \\ Ch^{\frac{1}{2}}\|e_{u,h}^{n,l}\|_{L^2} + Ch, & \text{for general } \gamma(s), k = 0 \text{ and } h = h_j. \end{cases} \quad (3.27)$$

Therefore the inequality (2.32) holds. Based on Theorem 1, we then obtain the error estimates as stated in the following theorem.

Theorem 3 Assume that the exact solution u of the ND problem (3.1) is smooth enough. Let $u_h^n \in V_h$ is the approximate solution obtained from the fully discrete scheme (3.9) based on a r th order TSIRK scheme. Then we have the following error estimates:

$$\max_{n\tau \leq T} \|u(\cdot, t_n) - u_h^n\|_{L^2} \leq \begin{cases} C(\delta)h^{k+1} + C\tau^r, & \text{for } s\gamma(s) \in L^1_{loc} \text{ and } \delta \text{ is fixed,} \\ C(h^k + \tau^r), & \text{for general } \gamma(s) \text{ and } k \geq 1, \\ C(h^{\frac{1}{2}} + \tau^r), & \text{for general } \gamma(s), k = 0 \text{ and } h = h_j, \end{cases} \quad (3.28)$$

where C is a constant independent of δ as $\delta \rightarrow 0$ in the estimates for the last two cases with a general $\gamma(s)$. The last two error estimates also indicate the approximate solution u_h would uniformly converge to the exact solution when $\tau, h \rightarrow 0$, regardless of choices of $\gamma(s)$ and δ , i.e., the fully discrete scheme (3.9) is asymptotically compatible.

Since the exact solution of the nonlocal diffusion problem is dissipative, we expect the L^2 -norm of the numerical solution decreases in time. In fact, u_h^n converges to a constant (with the periodic boundary condition) when $n \rightarrow \infty$ and h is fixed.

Proposition 1 *The long time asymptotic behavior of the approximate solution $u_h^n \in V_h$ produced by the fully discrete scheme (3.9) based on a TSIRK is as follows:*

$$\lim_{n \rightarrow \infty} \left\| u_h^n - \frac{1}{b-a} \int_a^b u_h^0(x) dx \right\|_{L^2} = 0. \quad (3.29)$$

Proof We only give a sketchy proof here since the proof is similar in [14]. With Conditions (H1) and (H2) both hold, we know that $u_h^n, u_h^{n,l}, l = 1, \dots, J$ are uniformly bounded in L^2 . Therefore, by the finite dimensional nature of V_h , there exists a subsequence $\{u_h^{n_j}\}_{j=1}^\infty$ s.t.

$$u_h^{n_j}(x) \rightarrow u_*(x).$$

Thus there exists $\{n_{j_k}\}_{k=1}^\infty$ s.t.

$$u_h^{n_{j_k},l} \rightarrow u_{*,l}(x), \quad \text{as } k \rightarrow \infty,$$

for some $u_*(x), u_{*,l}(x) \in V_h, l = 1, \dots, J$. Then let us define $q_{*,l}(x; s) \in V_h, l = 1, \dots, J$ satisfying

$$(q_{*,l}(\cdot; s), v_h) = \mathcal{H}(u_{*,l}, v_h; s), \quad \forall v_h \in V_h.$$

Since $u_h^{n_{j_k},l} \rightarrow u_{*,l}, l = 1, \dots, J$ as $k \rightarrow \infty$, we can obtain

$$q_h^{n_{j_k},l}(x; s) \rightarrow q_{*,l}(x; s), \quad \text{as } k \rightarrow \infty, \quad l = 1, \dots, J.$$

Let $n = n_{j_k}$ and $k \rightarrow \infty$ in (2.8), we obtain

$$q_{*,1}(x; s) = \dots = q_{*,J}(x; s) = 0 \quad \text{on } [a, b] \times (0, \delta).$$

This indicates

$$u_*(x) = u_{*,1}(x) = \dots = u_{*,J}(x) = \text{constant}, \quad \forall x \in [a, b].$$

And since the fully discrete scheme (3.9) preserves the cell averages, we obtain

$$u_*(x) = \frac{1}{b-a} \int_a^b u_h^0(y) dy,$$

which implies that $u_*(x)$ is unique and independent of the choice of the subsequence. Thus the whole sequence converges in the L^2 -norm sense, as in (3.29). $\square$

Table 4 L^2 errors and convergence orders with $\alpha = 1/2$ in Example 1

	N	Fixed δ				Fixed δ/h			
		$\delta = 10^{-12}$		$\delta = 1/4$		$\delta = h$		$\delta = 3h$	
		L^2 error	Order	L^2 error	Order	L^2 error	Order	L^2 error	Order
$\mathcal{P}_0$, BE	16	8.67E-02	–	8.61E-02	–	8.50E-02	–	8.64E-02	–
	32	4.29E-02	1.02	4.28E-02	1.01	4.27E-02	1.00	4.28E-02	1.01
	64	2.14E-02	1.00	2.14E-02	1.00	2.13E-02	1.00	2.14E-02	1.00
	128	1.07E-02	1.00	1.07E-02	1.00	1.07E-02	1.00	1.07E-02	1.00
	256	5.34E-03	1.00	5.34E-03	1.00	5.34E-03	1.00	5.34E-03	1.00
	512	2.67E-03	1.00	2.67E-03	1.00	2.67E-03	1.00	2.67E-03	1.00
$\mathcal{P}_1$, SDIRK(2,1)	16	1.35E-02	–	8.31E-03	–	8.36E-03	–	8.28E-03	–
	32	3.37E-03	2.00	2.07E-03	2.00	2.09E-03	2.00	2.08E-03	2.00
	64	8.43E-04	2.00	5.17E-04	2.00	5.21E-04	2.00	5.18E-04	2.00
	128	2.11E-04	2.00	1.29E-04	2.00	1.30E-04	2.00	1.29E-05	2.00
	256	5.27E-05	2.00	3.23E-05	2.00	3.25E-05	2.00	3.24E-06	2.00
	512	1.32E-05	2.00	8.07E-06	2.00	8.13E-06	2.00	8.09E-06	2.00
$\mathcal{P}_2$, SDIRK(3,2)	16	8.32E-04	–	5.49E-04	–	5.49E-04	–	5.50E-04	–
	32	1.05E-04	2.99	6.93E-05	2.99	6.93E-05	2.99	6.93E-05	3.00
	64	1.31E-05	3.00	8.68E-06	3.00	8.68E-06	3.00	8.68E-06	3.00
	128	1.64E-06	3.00	1.09E-06	3.00	1.09E-06	3.00	1.09E-06	3.00
	256	2.05E-07	3.00	1.36E-07	3.00	1.36E-07	3.00	1.36E-07	3.00
	512	2.56E-08	3.00	1.70E-08	3.00	1.70E-08	3.00	1.70E-08	3.00
$\mathcal{P}_2$, GL(4,2)	16	8.32E-04	–	5.49E-04	–	5.49E-04	–	5.50E-04	–
	32	1.05E-04	2.99	6.93E-05	2.99	6.93E-05	2.99	6.93E-05	2.99
	64	1.31E-05	3.00	8.68E-06	3.00	8.68E-06	3.00	8.68E-06	3.00
	128	1.64E-06	3.00	1.09E-06	3.00	1.09E-06	3.00	1.09E-06	3.00
	256	2.05E-07	3.00	1.36E-07	3.00	1.36E-07	3.00	1.36E-07	3.00
	512	2.56E-08	3.00	1.70E-08	3.00	1.70E-08	3.00	1.70E-08	3.00

Table 5 L^2 errors and convergence orders with $\alpha = 3/2$ in Example 1

	N	Fixed δ				Fixed δ/h			
		$\delta = 10^{-12}$		$\delta = 1/4$		$\delta = h$		$\delta = 3h$	
		L^2 error	Order	L^2 error	Order	L^2 error	Order	L^2 error	Order
$\mathcal{P}_0$, BE	16	8.67E-02	–	8.60E-02	–	8.53E-02	–	8.63E-02	–
	32	4.29E-02	1.02	4.28E-02	1.01	4.27E-02	1.00	4.28E-02	1.01
	64	2.14E-02	1.00	2.14E-02	1.00	2.14E-02	1.00	2.14E-02	1.00
	128	1.07E-02	1.00	1.07E-02	1.00	1.07E-02	1.00	1.07E-02	1.00
	256	5.34E-03	1.00	5.34E-03	1.00	5.34E-03	1.00	5.34E-03	1.00
	512	2.67E-03	1.00	2.67E-03	1.00	2.67E-03	1.00	2.67E-03	1.00
$\mathcal{P}_1$, SDIRK(2,1)	16	1.35E-02	–	8.75E-03	–	8.87E-03	–	8.71E-03	–
	32	3.37E-03	2.00	2.19E-03	2.00	2.25E-04	1.98	2.20E-04	2.01
	64	8.43E-04	2.00	5.47E-04	2.00	5.63E-04	2.00	5.52E-05	2.01
	128	2.11E-04	2.00	1.36E-04	2.00	1.41E-05	2.00	1.38E-05	2.01
	256	5.27E-05	2.00	3.40E-05	2.00	3.52E-06	2.00	3.45E-06	2.00
	512	1.32E-06	2.00	8.49E-06	2.00	8.81E-06	2.00	8.63E-06	2.00
$\mathcal{P}_2$, SDIRK(3,2)	16	8.32E-04	–	5.67E-04	–	5.71E-04	–	5.65E-04	–
	32	1.05E-04	2.99	7.08E-05	3.00	7.16E-05	2.99	7.09E-05	2.99
	64	1.31E-05	3.00	8.83E-06	3.00	8.96E-06	3.00	8.87E-06	3.00
	128	1.64E-06	3.00	1.10E-06	3.00	1.12E-06	3.00	1.11E-06	3.00
	256	2.05E-07	3.00	1.38E-07	3.00	1.40E-07	3.00	1.39E-07	3.00
	512	2.56E-08	3.00	1.72E-08	3.00	1.75E-08	3.00	1.73E-08	3.00
$\mathcal{P}_2$, GL(4,2)	16	8.32E-04	–	5.67E-04	–	5.71E-04	–	5.65E-04	–
	32	1.05E-04	2.99	7.08E-05	3.00	7.16E-05	2.99	7.09E-05	2.99
	64	1.31E-05	3.00	8.83E-06	3.00	8.96E-06	3.00	8.87E-06	3.00
	128	1.64E-06	3.00	1.10E-06	3.00	1.12E-06	3.00	1.11E-06	3.00
	256	2.05E-07	3.00	1.38E-07	3.00	1.40E-07	3.00	1.39E-07	3.00
	512	2.56E-08	3.00	1.72E-08	3.00	1.75E-08	3.00	1.73E-08	3.00

Table 6 L^2 errors and convergence orders with $\alpha = 5/2$ in Example 1

	N	Fixed δ				Fixed δ/h			
		$\delta = 10^{-12}$		$\delta = 1/4$		$\delta = h$		$\delta = 3h$	
		L^2 error	Order	L^2 error	Order	L^2 error	Order	L^2 error	Order
$\mathcal{P}_0$, BE	16	8.67E-02	–	8.61E-02	–	8.59E-02	–	8.62E-02	–
	32	4.29E-02	1.02	4.28E-02	1.01	4.28E-02	1.01	4.28E-02	1.01
	64	2.14E-02	1.00	2.14E-02	1.00	2.14E-02	1.00	2.14E-02	1.00
	128	1.07E-02	1.00	1.07E-02	1.00	1.07E-02	1.00	1.07E-02	1.00
	256	5.34E-03	1.00	5.34E-03	1.00	5.34E-03	1.00	5.34E-03	1.00
	512	2.67E-03	1.00	2.67E-03	1.00	2.67E-03	1.00	2.67E-03	1.00
$\mathcal{P}_1$, SDIRK(2,1)	16	1.35E-02	–	1.11E-02	–	1.11E-02	–	1.11E-02	–
	32	3.37E-03	2.00	2.79E-03	2.00	2.81E-03	1.99	2.79E-03	1.99
	64	8.43E-04	2.00	6.98E-04	2.00	7.05E-04	2.00	6.99E-04	2.00
	128	2.11E-04	2.00	1.74E-04	2.00	1.76E-04	2.00	1.75E-04	2.00
	256	5.27E-05	2.00	4.36E-05	2.00	4.41E-05	2.00	4.37E-05	2.00
	512	1.32E-05	2.00	1.09E-05	2.00	1.10E-05	2.00	1.09E-05	2.00
$\mathcal{P}_2$, SDIRK(3,2)	16	8.32E-04	–	6.81E-04	–	6.84E-04	–	6.80E-04	–
	32	1.05E-04	2.99	8.54E-05	3.00	8.59E-05	2.99	8.55E-05	2.99
	64	1.31E-05	3.00	1.07E-05	3.00	1.08E-05	3.00	1.07E-05	3.00
	128	1.64E-06	3.00	1.34E-06	3.00	1.34E-06	3.00	1.34E-06	3.00
	256	2.05E-07	3.00	1.67E-07	3.00	1.68E-07	3.00	1.67E-07	3.00
	512	2.56E-08	3.00	2.09E-08	3.00	2.10E-08	3.00	2.09E-08	3.00
$\mathcal{P}_2$, GL(4,2)	16	8.32E-04	–	6.81E-04	–	6.84E-04	–	6.80E-04	–
	32	1.05E-04	2.99	8.54E-05	3.00	8.59E-05	2.99	8.55E-05	2.99
	64	1.31E-05	3.00	1.07E-05	3.00	1.08E-05	3.00	1.07E-05	3.00
	128	1.64E-06	3.00	1.34E-06	3.00	1.34E-06	3.00	1.34E-06	3.00
	256	2.05E-07	3.00	1.67E-07	3.00	1.68E-07	3.00	1.67E-07	3.00
	512	2.56E-08	3.00	2.09E-08	3.00	2.10E-08	3.00	2.09E-08	3.00

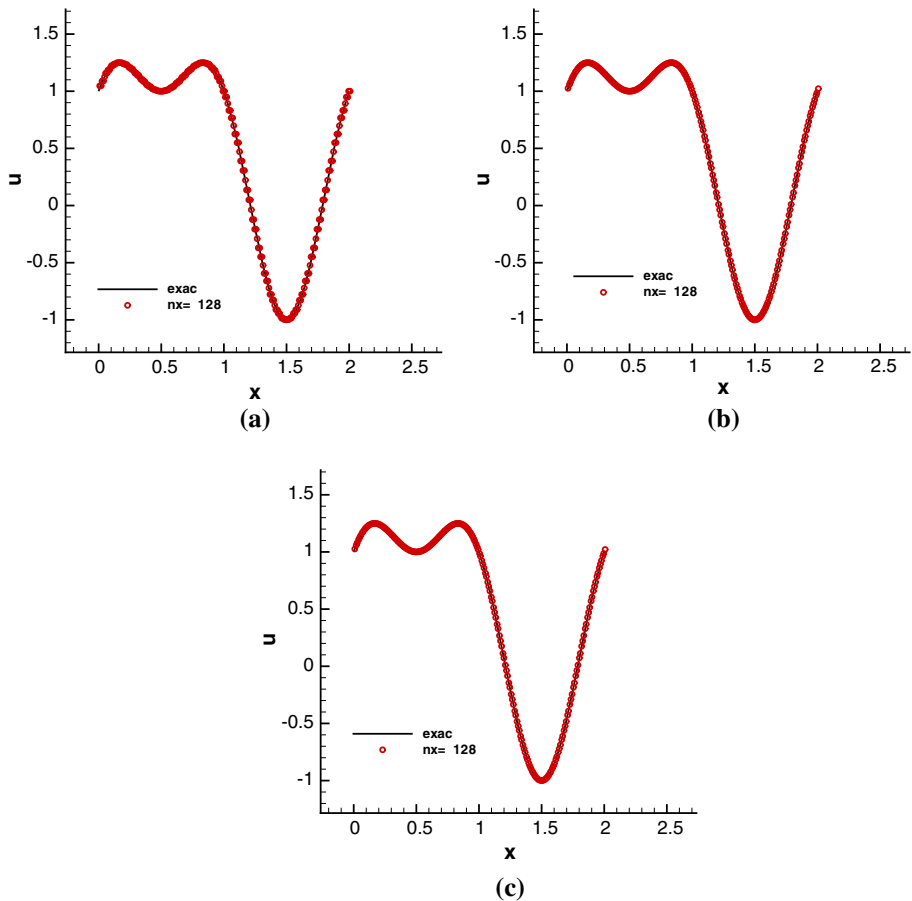


Fig. 1 Plots of the steady state solution u_h produced by the fully discrete scheme (3.9) with the spatial grid $N = 128$ in Example 2. **a** $\mathcal{P}_0$, BE, **b** $\mathcal{P}_1$, SDIRK(2,1), **c** $\mathcal{P}_2$, SDIRK(3,2)

4 Numerical Experiments

In this section we will perform some numerical experiments to verify the theoretical results. We set the kernel function as

$$\gamma(s) = \frac{3 - \alpha}{2\delta^{3-\alpha}} |s|^{-\alpha}, \quad 0 < \alpha < 3,$$

which gives us

$$\int_{-\delta}^{\delta} s^2 \gamma(s) ds = 1.$$

In addition, $\gamma(s)$ is integrable when $0 < \alpha < 1$. We adopt the fully discrete DG scheme (3.9) to solve the time-dependent ND problem (3.1). More specifically, the backward Euler (BE) method, SDIRK(2,1) (identically, GL(2,1)), SDIRK(3,2) and GL(4,2) are used for time stepping, that all satisfy Conditions (H1) and (H2). The domain $\Omega = (0, 2)$ is used, and we consider the uniform mesh, i.e. $h_j = h$. Also, we set $\tau = 0.1h$ and take $k = 0, 1, 2$

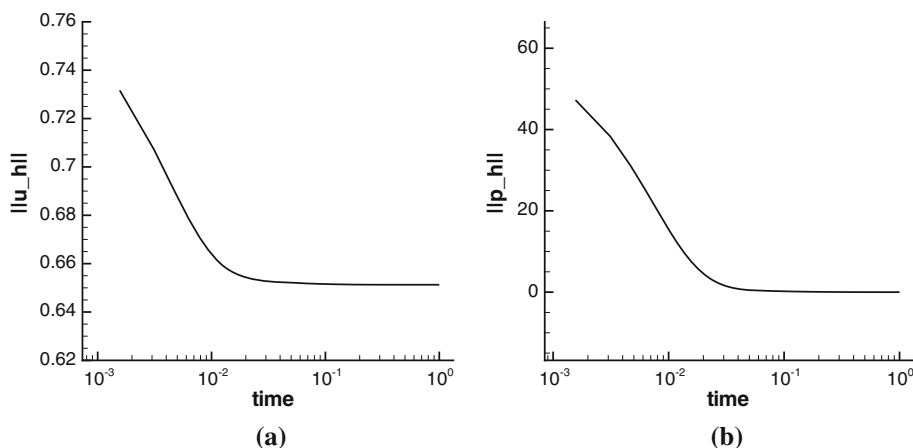


Fig. 2 Evolution of the $\|u_h^n\|_{L^2}$ and $\|p_h^n\|_{L^2}$ produced by the fully discrete scheme (3.9) with the spatial grid $N = 128$ in Example 3. **a** $\|u_h^n\|_{L^2}$ versus time, **b** $\|p_h^n\|_{L^2}$ versus time

for the order of the piecewise polynomials for spatial discretization ($V_h = \mathcal{P}_0, \mathcal{P}_1$ and $\mathcal{P}_2$, respectively). We treat the integral in (3.5) into two parts: $s \in (0, h)$ and $s \in (h, \delta)$. Since $s^2 \gamma(s)$ maybe singular around 0, for instance $\alpha > 2$, we compute the integral $s \in (0, h)$ exactly. For $s \in (h, \delta)$, we use the numerical quadrature with sufficient accuracy.

Example 1 We take an additional source function f_δ in the right hand side of Eq. (3.1) so that the exact solution is given by $u(x) = \cos(\pi x)^2 + e^{-t^2} \sin(\pi x)$. The final time is chosen to be $T = 1$.

We set the number of nodes in space $N = 16, 32, \dots, 512$, respectively, and present the simulation results in Tables 4, 5 and 6. From Tables 4, 5 and 6, we can clearly observe the optimal orders of convergence for the fully discrete scheme (3.9) when $\alpha = 1/2, 3/2, 5/2$. We can also see that the errors for SDIRK(3,2) are almost the same as that for GL(4,2) because the spatial errors are dominant in the case of GL(4,2), which are the third order of convergence.

Example 2 We next consider the asymptotic behavior of the fully discrete scheme after a long time evolution. The following additional source function $f_\delta(x)$ is added to the right hand side of Eq. (3.1):

$$f_\delta(x) = -2 \int_{-\delta}^{\delta} \gamma(s)(g(x+s) - g(x)) ds, \quad \text{where } g(x) = \cos(\pi x)^2 + \sin(\pi x).$$

The steady state solution of the problem (3.1) is then $u(x) = g(x)$. We take the initial condition $u_0(x) = 0$ and the stopping criterion is $\sum_j |\bar{u}_j^{n+1} - \bar{u}_j^n| \leq 10^{-6}$, where $\bar{u}_j^n$ is the cell average on I_j at time level t_n . The SDIRK method is adopted for time stepping.

We show the converged numerical solution u_h with the spatial grid of $N = 128, \alpha = 5/2$ and $\delta = 3h$ in Fig. 1, produced by the fully discrete scheme (3.9). We could see the numerical solution u_h always converges to the exact steady state solution after a long time evolution. We have tested several δ and α , and the numerical results show that u_h always converges nicely.

Example 3 In the last example we consider the dissipative phenomenon of the scheme (3.9). We take a random initial condition, and the u_h^0 is a constant on each cell I_j , within the range $(0,1)$. Then we study how the $\|u_h^n\|_{L^2}$ and $\|p_h^n\|_{L^2}$ decrease, where p_h^n is defined in Remark 4. We set $\alpha = 1/2$ and $\delta = 1/4$.

For simplicity, we only show the result with SDIRK(3,2) method in time discretization, and the degree of the polynomial is $k = 2$. We plot the evolution of $\|u_h^n\|_{L^2}$ and $\|p_h^n\|_{L^2}$ against the time in Fig. 2. It can be observed that both of them monotonically decrease, and $\|u_h^n\|_{L^2}$ finally converges to a positive constant along the time and $\|p_h^n\|_{L^2}$ to zero as expected.

5 Concluding Remarks

In this paper, we present stability and error analysis of the fully discrete numerical scheme for time-dependent problems. With a specific class of implicit Runge–Kutta time discretization, a TSIRK scheme that satisfies Conditions (H1) and (H2), and the spatial discretization satisfying the dissipative condition (2.3), we show the proposed fully discrete scheme is unconditionally strongly stable, and their numerical solution of intermediate stages are also stable. With extra hypotheses on the consistency errors of the discrete spatial operators, we obtain the priori error estimates for the fully discrete numerical scheme. As an application of this general framework, we then continue the study in [14] and propose some fully discrete numerical schemes for solving one-dimensional time-dependent nonlocal diffusion problems. Using the discontinuous Galerkin (DG) scheme for spatial discretization and a TSIRK time discretization, we verify the proposed fully discrete scheme is unconditionally strongly stable and derive the error estimates. The long time asymptotic behavior of the scheme is also investigated. Several numerical examples are tested to verify the theoretically results.

Still, there are a number of remaining issues. First, it will be helpful to see if the consistency error of spatial discretization can be more systematically studied. Moreover, one can naturally explore the application of the TSIRK framework to other linear or nonlinear dissipative physical systems. Obviously, concerning the nonlocal models, there are still many outstanding questions to be answered, for example, it will be interesting to see how to extend the current work on the one-dimensional nonlocal model problem to more complex and multi-dimensional problems.

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